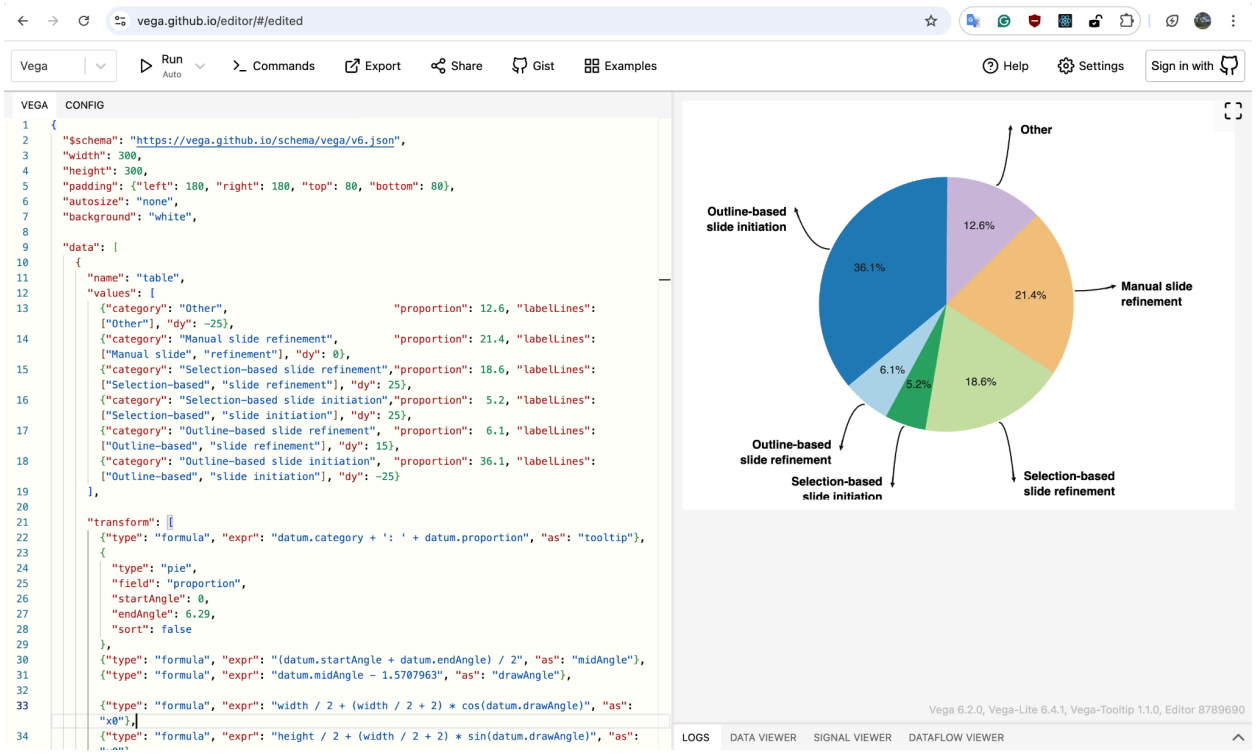
Task 1: Vega

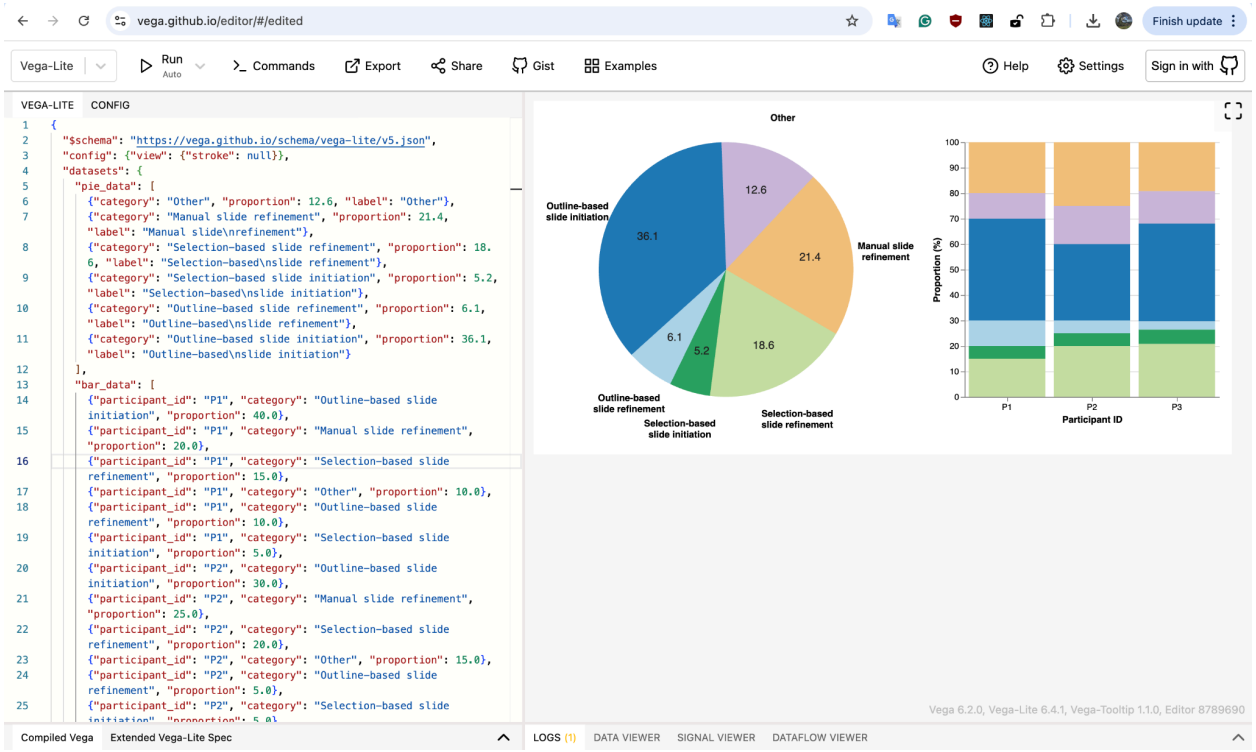


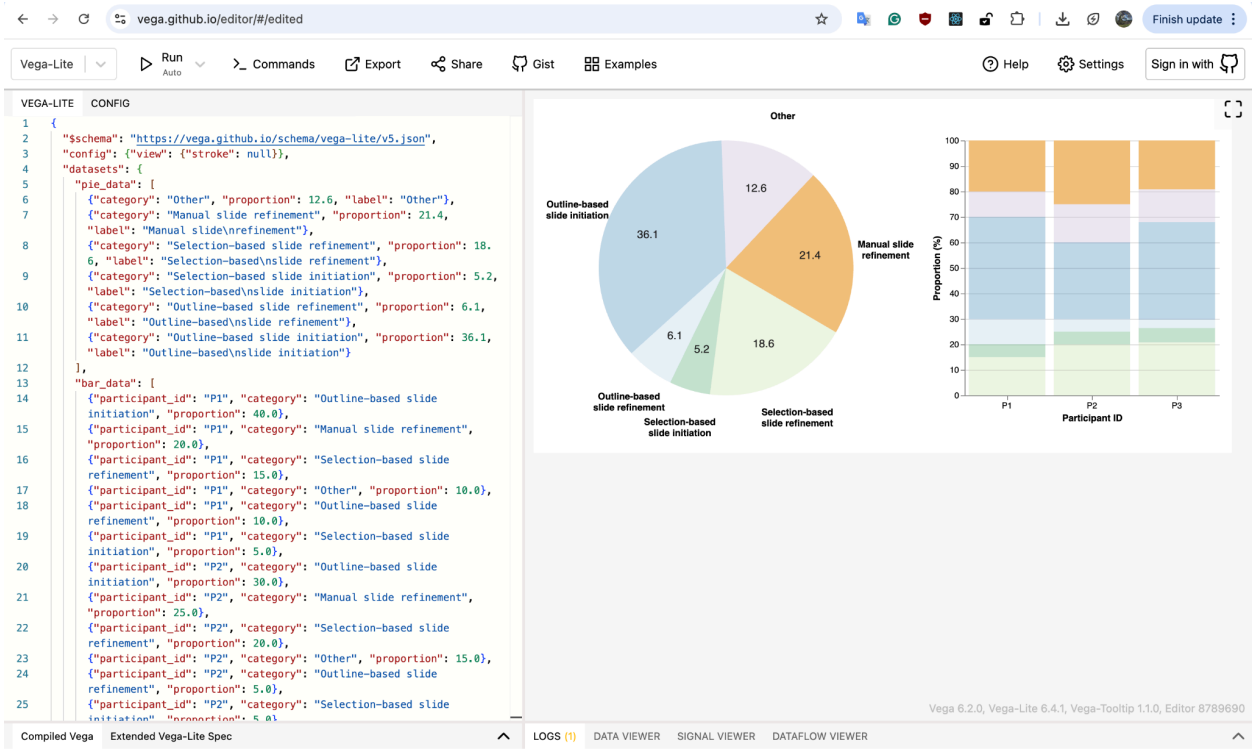
Task 1 Code

```
{
 "$schema": "https://vega.github.io/schema/vega/v6.json",
 "width": 300,
 "height": 300,
 "padding": {"left": 180, "right": 180, "top": 80, "bottom": 80},
 "autosize": "none",
 "background": "white",

 "data": [
  {
   "name": "table",
   "values": [
    {"category": "Other",                    "proportion": 12.6, "labelLines": ["Other"], "dy": -25},
    {"category": "Manual slide refinement",       "proportion": 21.4, "labelLines": ["Manual slide", "refinement"], "dy": 0},
    {"category": "Selection-based slide refinement","proportion": 18.6, "labelLines": ["Selection-based", "slide refinement"], "dy": 25},
    {"category": "Selection-based slide initiation","proportion":  5.2, "labelLines": ["Selection-based", "slide initiation"], "dy": 25},
    {"category": "Outline-based slide refinement",  "proportion":  6.1, "labelLines": ["Outline-based", "slide refinement"], "dy": 15},
    {"category": "Outline-based slide initiation",  "proportion": 36.1, "labelLines": ["Outline-based", "slide initiation"], "dy": -25}
   ],

   "transform": [
    {"type": "formula", "expr": "datum.category + ': ' + datum.proportion", "as": "tooltip"},
    {
     "type": "pie",
     "field": "proportion",
     "startAngle": 0,
     "endAngle": 6.29,
     "sort": false
    },
    {"type": "formula", "expr": "(datum.startAngle + datum.endAngle) / 2", "as": "midAngle"},
    {"type": "formula", "expr": "datum.midAngle - 1.5707963", "as": "drawAngle"},

    {"type": "formula", "expr": "width / 2 + (width / 2 + 2) * cos(datum.drawAngle)", "as": "x0"},
    {"type": "formula", "expr": "height / 2 + (width / 2 + 2) * sin(datum.drawAngle)", "as": "y0"},
    {"type": "formula", "expr": "width / 2 + (width / 2 + 45) * cos(datum.drawAngle)", "as": "x1"},
    {"type": "formula", "expr": "height / 2 + (width / 2 + 45) * sin(datum.drawAngle) + datum.dy", "as": "y1"},

    {"type": "formula", "expr": "datum.x0 + (datum.x1 - datum.x0) * 0.6", "as": "cx"},
    {"type": "formula", "expr": "datum.y0", "as": "cy"},
    {"type": "formula", "expr": "atan2(datum.y1 - datum.cy, datum.x1 - datum.cx) * 180 / PI + 90", "as": "arrowAngle"}
   ]
  }
 ],

 "scales": [
```

```
    {
      "name": "color",
      "type": "ordinal",
      "domain": [
        "Manual slide refinement",
        "Selection-based slide refinement",
        "Selection-based slide initiation",
        "Outline-based slide refinement",
        "Outline-based slide initiation",
        "Other"
      ],
      "range": ["#F4C07A", "#C5DFA0", "#2CA25F", "#AED6EA", "#1F77B4", "#C9B8D8"]
    }
  ],

  "marks": [
    {
      "type": "arc",
      "from": {"data": "table"},
      "encode": {
        "enter": {
          "fill": {"scale": "color", "field": "category"},
          "x": {"signal": "width / 2"},
          "y": {"signal": "height / 2"},
          "startAngle": {"field": "startAngle"},
          "endAngle": {"field": "endAngle"},
          "innerRadius": {"value": 0},
          "outerRadius": {"signal": "width / 2"},
          "cornerRadius": {"value": 0},
          "tooltip": {"field": "tooltip"}
        }
      }
    },
    {
      "type": "text",
      "from": {"data": "table"},
      "encode": {
        "enter": {
          "x":        {"signal": "width / 2 + (width / 2 - 50) * cos(datum.drawAngle)"},
          "y":        {"signal": "height / 2 + (width / 2 - 50) * sin(datum.drawAngle)"},
          "text":     {"signal": "datum.proportion + '%'"},
          "align":    {"value": "center"},
          "baseline":  {"value": "middle"},
          "fontSize":  {"value": 13},
          "fontWeight": {"value": "normal"},
          "fill":     {"value": "#222222"}
        }
      }
    },
    {
      "type": "path",
      "from": {"data": "table"},
      "encode": {
        "enter": {
          "path": {"signal": "'M' + datum.x0 + ',' + datum.y0 + ' Q' + datum.cx + ',' + datum.cy + ' ' + datum.x1 + ',' + datum.y1"},
          "stroke": {"value": "#111"},
          "strokeWidth": {"value": 1.5},
          "fill": {"value": "transparent"}
        }
      }
    },
    {
      "type": "symbol",
      "from": {"data": "table"},
      "encode": {
        "enter": {
          "x": {"field": "x1"},
          "y": {"field": "y1"},
          "shape": {"value": "arrow"},
          "size": {"value": 180},
          "fill": {"value": "#111"},
          "angle": {"field": "arrowAngle"}
        }
      }
    },
    {
      "type": "text",
      "from": {"data": "table"},
      "encode": {
        "enter": {
          "x":        {"signal": "datum.x1 + (cos(datum.drawAngle) >= 0 ? 12 : -12)"},
          "y":        {"field": "y1"},
          "text":     {"field": "labelLines"},
          "align":    {"signal": "cos(datum.drawAngle) >= 0 ? 'left' : 'right'"},
          "baseline":  {"value": "middle"},
          "fontSize":  {"value": 14},
          "fontWeight": {"value": "bold"},
          "lineHeight": {"value": 18},
          "fill":     {"value": "#000000"}
        }
      }
    }
  ]
}
```

Task 2: Vega-Lite





DataVis

task2.html U | visualization.vl.json

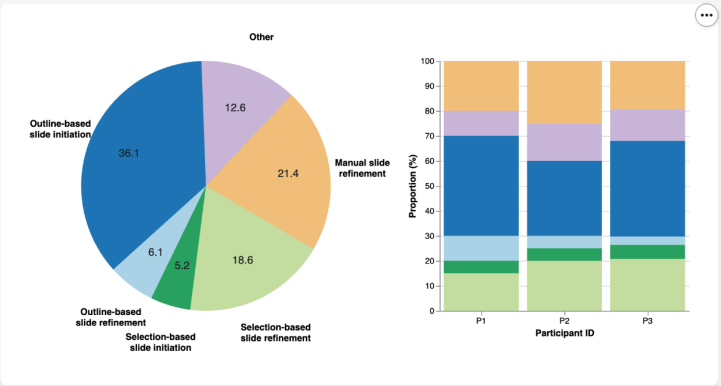
Lab6 > task2.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Interactive Vega-Lite Dashboard</title>

  <!-- Import Vega 5, Vega-Lite 5, and vega-embed 6 from a CDN -->
  <script src="https://cdn.jsdelivr.net/npm/vega@5"></script>
  <script src="https://cdn.jsdelivr.net/npm/vega-lite@5"></script>
  <script src="https://cdn.jsdelivr.net/npm/vega-embed@6"></script>

  <style>
    /* Basic styling to center the visualization on the page */
    body {
      font-family: sans-serif;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
      background-color: #f9f9f9;
    }
    #dashboard-container {
      background: white;
      padding: 30px;
      border-radius: 8px;
      box-shadow: 0 4px 10px rgba(0,0,0,0.1);
    }
  </style>
</head>
<body>
  <!-- The div where our Vega-Lite chart will be rendered -->
  <div id="dashboard-container"></div>

  <script type="text/javascript">
    // Assign your Vega-Lite JSON specification to a variable
    const vlSpec = {
      "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
      "config": {"view": {"stroke": null}},
      "datasets": {
        "pie_data": [
          {"category": "Other", "proportion": 12.6, "label": "Other"},
          {
            "category": "Manual slide refinement",
            "proportion": 21.4,
```

main* | Launchpad | 0 0 | Live Share | Update is ready, click to restart. | Ln 291, Col 1 | Spaces: 2 | UTF-8 | LF | { } HTML

127.0.0.1:5500/Lab6/task2.html

Finish update



Task 2 Code


```
{
 "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
 "config": {"view": {"stroke": null}},
 "datasets": {
  "pie_data": [
   {"category": "Other", "proportion": 12.6, "label": "Other"},
   {"category": "Manual slide refinement", "proportion": 21.4, "label": "Manual slide\nrefinement"},
   {"category": "Selection-based slide refinement", "proportion": 18.6, "label": "Selection-based\nslide refinement"},
   {"category": "Selection-based slide initiation", "proportion": 5.2, "label": "Selection-based\nslide initiation"},
   {"category": "Outline-based slide refinement", "proportion": 6.1, "label": "Outline-based\nslide refinement"},
   {"category": "Outline-based slide initiation", "proportion": 36.1, "label": "Outline-based\nslide initiation"}
  ],
  "bar_data": [
   {"participant_id": "P1", "category": "Outline-based slide initiation", "proportion": 40.0},
   {"participant_id": "P1", "category": "Manual slide refinement", "proportion": 20.0},
   {"participant_id": "P1", "category": "Selection-based slide refinement", "proportion": 15.0},
   {"participant_id": "P1", "category": "Other", "proportion": 10.0},
   {"participant_id": "P1", "category": "Outline-based slide refinement", "proportion": 10.0},
   {"participant_id": "P1", "category": "Selection-based slide initiation", "proportion": 5.0},
   {"participant_id": "P2", "category": "Outline-based slide initiation", "proportion": 30.0},
   {"participant_id": "P2", "category": "Manual slide refinement", "proportion": 25.0},
   {"participant_id": "P2", "category": "Selection-based slide refinement", "proportion": 20.0},
   {"participant_id": "P2", "category": "Other", "proportion": 15.0},
   {"participant_id": "P2", "category": "Outline-based slide refinement", "proportion": 5.0},
   {"participant_id": "P2", "category": "Selection-based slide initiation", "proportion": 5.0},
   {"participant_id": "P3", "category": "Outline-based slide initiation", "proportion": 38.3},
   {"participant_id": "P3", "category": "Manual slide refinement", "proportion": 19.2},
   {"participant_id": "P3", "category": "Selection-based slide refinement", "proportion": 20.8},
   {"participant_id": "P3", "category": "Other", "proportion": 12.8},
   {"participant_id": "P3", "category": "Outline-based slide refinement", "proportion": 3.3},
   {"participant_id": "P3", "category": "Selection-based slide initiation", "proportion": 5.6}
  ]
 },
 "hconcat": [
  {
   "width": 300,
   "height": 300,
   "data": {"name": "pie_data"},
   "encoding": {
    "theta": {
     "field": "proportion",
     "type": "quantitative",
     "stack": true
    },
    "order": {
     "field": "category",
     "type": "nominal",
     "sort": "descending"
    },
    "color": {
     "field": "category",
     "type": "nominal",
     "scale": {
      "domain": [
       "Other",
       "Manual slide refinement",
       "Selection-based slide refinement",
       "Selection-based slide initiation",
       "Outline-based slide refinement",
       "Outline-based slide initiation"
      ],
      "range": ["#C9B8D8", "#F4C07A", "#C5DFA0", "#2CA25F", "#AED6EA", "#1F77B4"]
     },
     "legend": null
    }
   },
   "layer": [
    {
     "params": [
      {
       "name": "category_select",
       "select": {"type": "point", "fields": ["category"]}
      }
     ],
     "mark": {
      "type": "arc",
      "outerRadius": 150,
      "thetaOffset": 2.1,
      "cursor": "pointer"
     },
     "encoding": {
      "opacity": {
       "condition": {"param": "category_select", "value": 1},
       "value": 0.3
      }
     }
    },
    {
     "mark": {
      "type": "text",
      "radius": 100,
      "fontSize": 13,
      "thetaOffset": 2.1
     },
```

```
      "encoding": {
        "text": {"field": "proportion", "type": "nominal", "format": ".1f"},
        "color": {"value": "#222"}
      }
    },
    {
      "mark": {
        "type": "text",
        "radius": 190,
        "fontWeight": "bold",
        "lineBreak": "\n",
        "thetaOffset": 2.1
      },
      "encoding": {
        "text": {"field": "label", "type": "nominal"},
        "color": {"value": "black"}
      }
    }
  ]
},
{
  "width": 300,
  "height": 300,
  "data": {"name": "bar_data"},
  "mark": "bar",
  "encoding": {
    "x": {
      "field": "participant_id",
      "type": "nominal",
      "title": "Participant ID",
      "axis": {"labelAngle": 0}
    },
    "y": {
      "field": "proportion",
      "type": "quantitative",
      "title": "Proportion (%)",
      "stack": "zero"
    },
    "order": {
      "field": "category",
      "type": "nominal",
      "sort": "descending"
    },
    "color": {
      "field": "category",
      "type": "nominal",
      "title": "Category",
      "scale": {
        "domain": [
          "Other",
          "Manual slide refinement",
          "Selection-based slide refinement",
          "Selection-based slide initiation",
          "Outline-based slide refinement",
          "Outline-based slide initiation"
        ],
        "range": ["#C9B8D8", "#F4C07A", "#C5DFA0", "#2CA25F", "#AED6EA", "#1F77B4"]
      }
    },
    "opacity": {
      "condition": {"param": "category_select", "value": 1},
      "value": 0.3
    }
  }
}
]
}
```


Task 3: Grammar Analysis

Vega-Lite is much more concise and provides rapid prototyping through high-level abstractions. For example, creating the pie layout and stacked bar chart only required simple *theta*, *color*, and *stack: "zero"* encodings, while cross-filtering between the pie and bar charts was achieved with a single *params* block. However, this abstracts away some control. In raw Vega (Task 1), we achieved higher expressiveness by explicitly defining SVG path strings for curved leader lines and calculating *atan2* formulas to precisely rotate arrowhead symbols.